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Denoising for balanced representation: A diffusion-informed approach to causal effect estimation

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ABSTRACT

Estimating individual treatment effects with observational data is crucial in causal inference. While it faces two major challenges: the absence of counterfactual outcomes and selection bias due to non-random assignment and intervention of confounding variables. This paper specifically addresses these by quantifying the distribution shift between the treated and control groups from a generative view to achieve balanced representation and then predict causal effects. We propose a novel method called **D**enoising for a **B**alanced **R**epresentation in **T**reatment Effect Estimation (**DBRT**). Motivated by the aim to generate analogous distributions in the Diffusion model, which aligns with our goal of achieving similar distributions in the latent space, our approach considers the divergence between the treated and control groups as'noise' that needs to be removed incrementally. We then construct a network to fit and then denoise it step by step, ensuring a consistent reduction of discrepancies between the two groups. Additionally, to enhance the representation of input features, we incorporate memory vectors into the attention mechanism to capture prior information and combined correlations between covariates. Furthermore, we utilize the Hilbert–Schmidt Independence Criterion (HSIC) within the loss function to constrain the learned representation, ensuring its relevance to the original data. DBRT has demonstrated superior performance on different datasets compared to previous classic works, highlighting its effectiveness in treatment effect estimation.

1. Introduction

Causal effect estimation refers to assessing the impact of specific treatment variable on outcomes [1]. It helps determine whether the intervention should be implemented in many domains like healthcare [2], economics [3], targeted advertising [4] and so on. The impact can be measured from individual and population levels, so we can obtain results for Individual Treatment Effects (ITE) and Average Treatment Effects (ATE) respectively. The estimated results can subsequently contribute to formulating customized policies and enabling more effective decision-making to maximize benefits. For instance, devising vocational training policies for relevant individuals can improve unemployment rates and stimulate economic development in social governance.

Randomized Controlled Trial (RCT) is regarded as the gold standard because treatments are randomly assigned to different groups. Then causal effect can be obtained by comparing the divergence of the group results. But considering RCT's high cost and potential ethical risk in practice, estimating from observational data is a good alternative [5]. This paper will focus on the problem of predicting ITE from observational data, which involves two main challenges.

Firstly, when an individual receives a specific treatment, only the outcomes under that treatment are observed, leaving the potential outcomes of other treatments unknown. It leads to the absence of counterfactuals. The fundamental basis for comparing treated and control outcomes is lacking, which makes it hard to get accurate treatment effects under the influence of confounding variables. Secondly, an individual's choice of treatment has been influenced by personal circumstances or preferences in observational data, resulting in the non-random assignment of interventions and distribution shift. This, along with the confounding variables can violate the assumptions of data independence and identical distribution [6]. It leads to the issue of selection bias. This means the treated and control groups differ systematically in both observed and unobserved ways, which may distort the prediction.

To address these challenges, researchers have proposed various strategies. Representation learning try to construct a latent space [7] that maps covariates of the treated and control groups to a balanced distribution [8]. It reduces the interference of confounders and enables

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more accurate outcome prediction [9]. Generative methods provide a new perspective as they can simulate the data generation process to further extract hidden proxy and identify implicit correlations. Leveraging this advantage, methods such as Variational Autoencoders (VAE) and Generative Adversarial Networks (GAN) have been employed to explore the hidden confounder [10] or predict counterfactual outcomes [11]. However, these methods with substantial parameters pose a higher risk of mode collapse during training [12,13]. The direct generation of new samples also has the potential to violate the assumptions of consistency [14], which could undermine the credibility of the results.

Therefore, we choose the diffusion probabilistic model, also known as Diffusion for short, for their strong theoretical foundation and stable training. Diffusion models recover the original distribution by progressively removing noise with each step in the reverse process. We draw inspiration from this approach to address selection bias. In our approach, we attempt to quantify the divergence between the treated and control groups from a generative view and mitigate these biases in the latent space. Our goal aligns with Diffusion's aim of generating analogous distributions. So we treat the divergence as the role of noise that has been introduced during the forward process in Diffusion. Then we model the noise through a specific network and gradually denoise it over several time steps to make the distributions more consistent.

Meanwhile, a good representation will explore the complex interactions among multiple factors of input covariates [15]. Previous works have mainly relied on fully connected neural network for representation, such as [16] [17], which may overlook the combined effects and thus miss some hidden information [18]. To explore the extra information, we incorporate memory vectors into the attention mechanism as the representation module. The additional memory vector integrates prior knowledge of the data and the attention mechanism learn relationship of covariates. Furthermore, we utilize Hilbert–Schmidt Independence Criterion (HSIC) within the loss function to constrain the relevance of representation to original covariates. Notably, HSIC is apt for detecting non-linear relationships by using kernel functions [19,20]. By combining the memory-augmented attention mechanism and HSIC as the loss function, our approach significantly enhances information representation.

Motivated by the challenges and premise mentioned above, we propose a novel method called **D**enoising for a **B**alanced **R**epresentation in **T**reatment Effect Estimation (**DBRT**), aiming to eliminate distribution shift for accurate individual effect estimation. It is based on the idea of progressive denoising from a causal generative perspective and the attention mechanism combined with memory vector for in-depth data mining. The main contributions are as follows:

- We consider the divergence between the treated and control group as noise and construct a specific network to fit it. Then the distribution shift will be stably removed after several timesteps. It aims for balanced distribution motivated by the reverse process in Diffusion.
- We incorporate memory vectors into the attention mechanism, enabling the model to learn prior knowledge of the data and capture the combined correlations between covariates. Thus it leads to more discriminative representations and wide generalization.
- We adopt the Hilbert–Schmidt Independence Criterion (HSIC) as the loss function for representation learning. It constrains the model to learn better embeddings from original data.
- We validate our method through experiments and visualization of the results on different datasets, which demonstrates the superiority of our approach.

To our knowledge, this study is the first attempt to estimate counterfactual outcomes by integrating representation learning and generative modeling within the framework of diffusion model. It reveals valuable insights into selection bias and paves the way for more robust results by mitigating the distribution shift stably. We will present our proposed methodology in the following aspects. In Section 2, we provide the related work of the direction covered by this paper. The problem setup will be elaborated in Section 3. Section 4 focuses on a detailed description of our proposed method. Implementation details, datasets, evaluation metrics, and experimental results will be presented in Section 5. Finally, we will conclude the entire paper in Section 6.

2. Related work

In recent years, researchers have proposed various methods to address the obstacles in ITE estimation from different perspectives. Reweighting methods [21] adjust sample weights according to different designs of balancing scores to make the distributions similar and alleviate the bias. Matching methods [22] aim to find nearest neighbors in the opposite group to infer counterfactual outcomes. Tree-based methods [23,24] learn decision rules through Bayesian additive tree, causal random forest and others to regress the results. But these may still be insufficient to discern hidden confounders and eliminate its adverse effects when faced with high-dimensional data or complex covariate relationships. Deep learning methods appear to be a good choice.

Representation balancing: map covariates into embeddings $\phi(x)$ to make distributions of $\phi(x|t=0)$ and $\phi(x|t=1)$ similar. It is similar to the problem of domain adaption. BNN [25] was the first to apply this approach to counterfactual inference and constructed a simple network as the latent encoder. Then TarNet [16] improved it by structing a two-head network with shared layers. The design can learn both common and distinct information of covariates. Based on it, CFRNet [16] utilized integral probability metrics to bound the distance between representation spaces. To maintain local similarity in representation, SITE [17] employed triplet pairs selection and strategies of position-dependent deep metric (PDDM) and middle point distance minimization (MPDM). Moreover, SCI [26] partitioned covariates into several subspaces and combined these information after balancing for better performance. CITE [27] integrates contrastive tasks specifically designed for causal inference, effectively utilizing latent self-supervised information to generate balanced, predictive representations while leveraging causal prior knowledge.

Different from previous state-of-the-art methods, our approach introduces an innovative perspective by considering the bias between the treated and control groups as noise. Then we revolutionize the way to achieve representation balance by constructing a specific network to fit the noise and then denoise it step by step.

Generative causal learning: employs generative modeling approaches for causal discovery and causal effect estimation [28]. In the task of causal discovery, generative models are typically employed to learn the distribution of data to capture the underlying causal mechanisms of data. In the task of causal effect estimation, generative models approximate data distributions to capture underlying patterns. Thereby it can further achieve balanced representations that mitigate confounder effects or directly produce counterfactual outcomes. CEVAE [10] utilized VAE to identify proxy confounders and then minimized their impact to infer outcome. CEIGM [29], on the other hand, leverages implicit generative models to tackle the challenge of hidden confounders by simulating the data generation process through GAN. GANITE [11] introduced GAN into causal inference. It leveraged adversarial learning to estimate potential outcomes from factual data. Subsequent works such as SCIGAN [30] improved it by using a hierarchical discriminator, and GAD [31] applied GAN to continuous treatment. ABCEI [32] utilized GAN to achieve representation balancing with the constraints of mutual information to alleviate representation loss. CBRE [33] furthered this idea by introducing cycle consistency to preserve semantic information and address the issue of information loss during the transformation of data into the latent space.

However, VAE and GAN might be caught in an unstable dilemma due to a large number of parameters in training. We attempt to addresses this limitation by employing diffusion model [34]. The rigorous

mathematical derivations of diffusion model enhance stability as well as provide stronger interpretability. It has been commonly employed in computer vision tasks, including image generation, restoration and so on [35] in recent years. DCM [36] proposed to generate unique latent encodings with diffusion to learn causal mechanisms under structural causal models(SCM). To the best of our knowledge, our study pioneers the application of diffusion model under potential outcome framework(POF) in causal inference.

Transformer in causal: Transformer has achieved tremendous success particularly in the domains of image and text. It also attracted researchers to explore its application in causal. Causal-Transformer [37] combined three transformer subnetworks and integrated them into a joint network with cross-attentions to capture long-term dependencies of temporal data. While there were limited researches on the application of Transformer in static data. CETransformer [38] utilized attention mechanism to represent the correlation between covariates. TansTEE [39] further incorporated treatment variables into the network.

We want to extend our approach by integrating prior knowledge of covariates into the network to enhance representation. In the field of computer vision, Meshed-Memory Transformer [10] introduced memory "slots" to learn prior knowledge about relationships between image regions. It has outstanding performance in image caption tasks. Taking advantage of this, we extend its Memory-Augmented Attention to causal inference for the first time to learn comprehensive representation for covariates and employ HSIC to constrain the representation.

In addition to the three paradigms mentioned above, there are some unique network designs. For instance, HTENet [40] used a multi-task neural network for direct treatment effect estimation, effectively avoiding issues related to model misspecification and high variability. ACIQR [41] reframed it as a quantile regression task within a bi-level optimization scheme, enabling it to handle diverse data structures. However, these recent methods do not thoroughly address the imbalance issue within the distribution shift that we aim to solve.

3. Preliminaries

3.1. Problem setup

Our work follow potential outcome framework proposed by [42]. Specifically, we suppose that the observational data consists of n units of $\{X_\mathrm{i}, T_\mathrm{i}, Y_\mathrm{i}\}$. In this set, $X_\mathrm{i} \in \mathbb{R}^{1\times k}$ means a covariate vector of each unit with k features. $T_\mathrm{i} \in \{0,1\}^n$ means treatment assignment. If $T_i = 1$, the unit belongs to the treated group, or to the control group if $T_i = 0$. $Y_i = Y(T_i) = T_iY_i(1) + (1-T_i)Y_i(0)$ means the factual outcome under the corresponding treatment. Then counterfactual outcome is $Y(1-T_i)$, which does not actually occur. The individual treatment effect (ITE) calculates the divergence between the potential treated and control for each unit i, which is defined as the $ITE(i) = Y_i(1) - Y_i(0)|X_i = x$. While $ATE = Y(1) - Y(0)$ denotes the gap between the two groups. Obviously only one of $Y_i(1)$ and $Y_i(0)$ is known, so we should handle the absence of counterfactual outcome. We aim at establishing a generalized model to estimate the ITE accurately by minimizing the prediction loss of potential outcomes.

3.2. Assumption

We intend to establish a reliable model to reduce selection bias based on the idea of representation balancing. So we need to satisfy the following assumptions to predict the causal effect with the original inputs.

SUTVA: Stable Unit Treatment Value Assumption (SUTVA) means the outcome of an individual under a specific treatment is independent of the others. It ensures the stability of the computed results.

Positivity: can be defined as $0 < p(T|X, Y(0), Y(1)) < 1$. It ensures that units have chance to receive each of the treatments.

Algorithm 1 Pseudo-code of DBRT

Input: Factual observational data $\{X_\mathrm{i}, T_\mathrm{i}, Y_\mathrm{i}\}$; the representation encoder φ with parameter θ_φ; the denoising network z with parameter θ_z, the decoder ψ with parameter θ_ψ; outcome prediction network f with parameter θ_f

Parameters: $\{\theta_\varphi, \theta_\psi, \theta_z, \theta_f\} = Random\ Initialize()$

Output: ITE prediction

1: Sample data into batches and get $\{\mathrm{x_i}, t_\mathrm{i}, y_\mathrm{i}\}$.

2: Map original covariates $\mathrm{x_i}$ into the latent space through the Memory-Augmented Encoder φ to get the treated, control and common representation $Z_{tre} = \varphi(x_{tre}, \vartheta)$, $Z_{con} = \varphi(x_{con}, \vartheta)$ and $Z_{com} = \varphi(x, \vartheta)$.

3: Denoise the noise $z_\theta(x_t, t)$ between Z_{tre} and Z_{con} progressively and get a balanced distribution of $\tilde{Z}_{con} = Z_{con} - \sum_{n=0}^{t} z_\theta(x_n, t_n)$ with Z_{tre}.

4: Map the representation Z_{tre} and Z_{con} back to the original dimension to get $\tilde{X}_{tre}$ and $\tilde{X}_{con}$ through ψ.

5: Concatenate each representation of Z_{tre} and Z_{con} with the common one Z_{com} to predict the outcome f_{tre} and f_{con}.

6: Estimate the individual treatment effect ITE from predicted f_{tre} and f_{con} and minimize

Loss functions: $\mathcal{L}_{recon}$, $\mathcal{L}_{diff}$, $\mathcal{L}_{pre}$

Set $\theta_\varphi, \theta_\psi \leftarrow Maximize\left(\mathcal{L}_{\mathrm{recon}}(\tilde{X}, X_i), \theta_\varphi, \theta_\psi\right)$

Set $\theta_z, \theta_\varphi \leftarrow Minimize\left(\mathcal{L}_{diff}(Z_{tre}, \tilde{Z}_{con}), \theta_z, \theta_\varphi\right)$

Set $\theta_f, \theta_\varphi \leftarrow Minimize\left(\mathcal{L}_{pred}(f_i, y_\mathrm{i}), \theta_f, \theta_\varphi\right)$

Ignorability: can be defined as $\{Y(0), Y(1)\} \perp T|X$. It means that there are no other hidden confounding variables between treatment assignment and potential outcomes. In other words, it ensures that our model is not disturbed by other factors.

Consistency: ensures that potential outcomes are correctly assigned to their corresponding treatment assignments.

4. Methodology

This section will introduce the proposed Denoising for a Balanced Representation in Treatment Effect Estimation (DBRT) model. As shown in Fig. 1, the model consists of four main modules: (1) an encoder $\varphi : \mathcal{X} \to \mathcal{Z}$utilizing Memory-Augmented Attention to map the covariates to latent embeddings; (2) a denoising module based on simplified U-Net architecture for balance between representation ; (3) a decoder $\psi : \mathcal{Z} \to \tilde{X}$ that maps the representation back to the original dimension and employs Hilbert–Schmidt Independence Criterion (HSIC) to calculate the reconstruction loss; (4) a prediction network $F : \mathcal{Z} \times T \to Y$ that estimates the potential outcomes based on the learned representation. The overall combination of these four modules improves prediction ability of potential outcomes by eliminating distribution shift.

The pseudo-code of Denoising for a Balanced Representation in Treatment Effect Estimation (DBRT) is listed in Algorithm 1. It presents the overall process of our method. Then, we will detail each module and training procedure of DBRT as follows.

4.1. Memory-augmented encoder

The issue of balancing the representation of the treated group and control group is equivalent to domain adaptation problem to some extent. [43] mentioned that constructing an appropriate latent space is crucial for the solution of domain adaption. To amplify the information of covariates in observational data, we attempt to design an ideal encoder which can explore the correlations between the features. **The attention mechanism effectively captures implicit information among covariates.** The attention mechanism calculates the

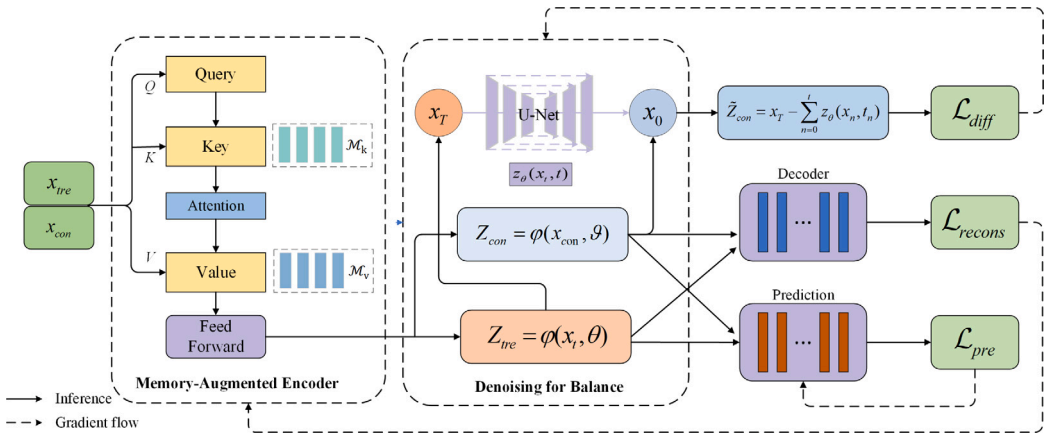

Fig. 1. Overview of Denoising for a Balanced Representation in Treatment Effect Estimation (DBRT). x_{tre} and x_{con} are the inputs, which denote covariates from treated and control group. M_k and M_v are additional memory vector to learn prior knowledge. Z_{tre} and Z_{con} are representations of x_{tre} and x_{con}, while $\tilde{Z}_{con}$ means the balanced representation after denoising network. $\mathcal{L}_{diff}$, $\mathcal{L}_{recon}$ and $\mathcal{L}_{pre}$ are loss functions for the model.

relevance of each vector to the context and then assigns corresponding importance weights, which is defined as:

$$Attention(Q, K, V) = soft\max\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{1}$$

where Q, K, V are the query, key and value vector obtained by linearly projecting original input [44], d is their dimension. However, this self-attention mechanism primarily focuses on pairwise relationships and may pay less attention to the global relationships among all components. This results in focusing on individual variable impacts while overlooking their combined effects. It is often the complex interactions among multiple factors that contribute to the final outcome in actual practice. Thus, incorporating prior knowledge can enhance the understanding of these interactions. Moreover, the number of covariates in our causal effect estimation task is much smaller compared to common scenarios in image and text. In order to enhance information representations and prevent overfitting, we use an improved mechanism called Memory-Augmented Attention, which is defined as:

$$\begin{aligned} &\mathcal{M}_{mem}(X) = Attention(W_q X, K, V) \\ &K = [W_k X, M_k] \\ &V = [W_v X, M_v] \end{aligned} \tag{2}$$

where M_k and M_v are additional vector which extend the original key and value. W_q, W_k, W_v are learnable weights of query, key and value vectors. Incorporating extra key and value vectors allows for learning prior knowledge not embedded in covariates, which will potentially overcome limitations. The memory vector is automatically updated through learnable matrices instead of depending on supplementary inputs. Thus, we utilize the Memory-Augmented Attention in the encoder to enrich the understanding of covariates and lead to a more comprehensive representation. To distinguish the treated group and control group, we construct two separate latent representation spaces for each. We also construct a common latent representation space for all covariates from the two groups. It will be jointly inputted with the separate subspace into the subsequent denoising process to enhance the overall understanding of the data. Through the mapping of these spaces, original covariates will become the embeddings of $Z_{tre} = \varphi(x_{tre}, \vartheta)$, $Z_{con} = \varphi(x_{con}, \vartheta)$ and $Z_{com} = \varphi(x, \vartheta)$,where x are all covariates regardless of treated or not, x_{tre} and x_{con} are the covariates from treated and control groups, ϑ is the parameter.

4.2. Denoising for balanced representation

The characteristics of observational data cause the presence of distribution shift, which, in turn, will interfere with estimation of results [9]. Therefore, we propose a method by considering the bias between the treated group and control group as noise and remove it stably with reference to the reverse course in diffusion model.

The diffusion probabilistic model [34] is based on the idea of a Markov chain, where Gaussian noise is gradually added to the original data in the forward process from x_{t-1} to x_t until x_T. x_{t-1}, x_t means the data of adding noise on the $t-1^{th}$th and tth timestep, and x_T is the final result. While in the reverse process, a model will be trained to learn and fit noise under the principle of maximum likelihood estimation. By removing the predicted noise progressively from x_t to x_{t-1} until $\tilde{x}_0$, it aims to generate samples that closely resemble the original data. $\tilde{x}_0$ means the denoised result after T timesteps. The denoising course is defined as:

$$p_\theta(x_{t-1} \mid x_t) := \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \tag{3}$$

where $\mu_\theta(x_t, t)$ and $\Sigma_\theta(x_t, t)$ are the mean and variance. Through variational inference and parameterization, the former can be expanded as:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{a_t}}(x_t - \frac{\beta_t}{\sqrt{1-\bar{a}_t}} z_\theta(x_t, t)) \tag{4}$$

where $z_\theta(x_t, t)$ is the noise predicted by the covariate distribution x_t at each timestep t. a_t, β_t and $\bar{a}_t$ are simplified to constants in our method.

As mentioned above, the reverse process only requires estimating noise at each time step and removes it step by step in diffusion model to approximate the original distribution of the data. This approach uses fewer parameters and offers greater stability compared to directly fitting the entire distribution, compared to VAE and GAN [45]. Additionally, Cold Diffusion [46] has demonstrated that transformations such as blur and pixelation in the forward process, as an alternative to Gaussian noise, can also lead to satisfactory performance. Motivated by this, we apply the reverse process to eliminate the divergence between the treated and control group.

Then we consider the representation of the treated group Z_{tre} as the targeted distribution x_0 and accordingly the representation of the control group Z_{con} as x_T, which is the mixture result of x_0 and selection bias. We set a total of T timesteps to progressively predict the noise $z_\theta(x_t, t)$ and remove it to get the transformed distribution of control group $\tilde{Z}_{con}$ at each timestep $n\,(0 < n < T)$. $\tilde{Z}_{con}$ become similar to Z_{tre}

after T timesteps' denoising, thereby it achieves the goal of eliminating bias caused by confounding variables. The course can be defined as:

$$\tilde{Z}_{con} = \tilde{x}_0 = x_T - \sum_{n=0}^{T} z_\theta(x_n, t_n) \tag{5}$$

We construct a specific network to predict and fit the noise $z_\theta(x_t, t)$. After referencing the classic U-Net [47], we simplified layers and dimensions to suit our data. We leverage the encoder–decoder structure with skip connections, so it is able to capture both global and detailed information of the features. Moreover, this network architecture employs fewer parameters, which is suitable for our task with limited features to prevent overfitting.

When evaluating the fit of noise $z_\theta(x_t, t)$, we adopt the Wasserstein distance as the integral probability metric (IPM) to measure the gap between denoised representation $\tilde{Z}_{con}$ and original representation Z_{tre}. Based on the theory of optimal transport, it can measure the similarity between two probability distributions and provide meaningful gradient during optimization even for sparse parts [48]. We define the loss function as:

$$L_{diff} = Wass\left(\tilde{Z}_{con}, Z_{tre}\right) = \inf_{\gamma \in \Pi[\tilde{Z}_{con}, Z_{tre}]} \iint \gamma(\theta, t) d(\theta, t) \mathrm{d}\theta \mathrm{d}t \tag{6}$$

where $Wass$ refers to the calculation of Wasserstein distance.

4.3. HSIC for loss reconstruction

After the transformation through the encoder $\varphi : \mathcal{X} \to \mathcal{Z}$ mapping covariates into a latent space, we construct a decoder $\psi : \mathcal{Z} \to \tilde{X}$ to map the embeddings back to the original dimension. The decoder is modeled by a feedforward neural network. Then we will get $\tilde{X}_{tre}$ and $\tilde{X}_{con}$ from the representation of Z_{tre} and Z_{con} after reconstruction. Previous methods applied Mean Squared Error (MSE) [38] or mutual information [32] to evaluate the loss between the transformed representation and original covariates. While MSE calculates the Euclidean distance between labels and predictions, which is not aligned with the goal of assessing whether the representation can capture the information of original data. Similarly, mutual information involves estimating the joint probability distribution of whole data, which may be challenging in many circumstances.

In contrast, the Hilbert–Schmidt Independence Criterion (HSIC) quantifies the interdependence between two sets of variables through the expectation of inner products of the kernel function from data samples directly. The kernel function is able to measure the nonlinear correlation between variables. This makes HSIC suitable for capturing complex relationships after the transformation through attention mechanism. Furthermore, the non-parametric property of HSIC removes the need for additional assumptions, making it flexible for various data distributions. Therefore, we employ HSIC to measure the loss during representation due to its suitability and efficiency. HSIC can be defined as:

$$HSIC(X, Y) := \frac{1}{n(n-1)} Tr(K_X H K_Y H) \tag{7}$$

where X, Y are the distributions to estimate correlations, n is their number of distribution, $H = I_n - \frac{1}{n} 1_{n\times n}$ means a kernel centering matrix, $K_X, K_Y \in \mathbb{R}^{n\times n}$ mean the kernel matrices of X and Y. The kernel matrices are able to measure the nonlinear correlation between variables [19]. We design the loss function as follows:

$$L_{recon} = \max(HSIC(X_{tre}, \tilde{X}_{tre}) + HSIC(X_{con}, \tilde{X}_{con})) \tag{8}$$

In the process of model optimization, L_{recon} converges towards a high correlation between the learned representation $\tilde{X}_{tre}$, $\tilde{X}_{con}$ and the original data X_{tre}, X_{con}, thereby ensuring constrained effects.

4.4. Outcome prediction

Following the modules introduced above, we address the issue of distributional shift and then we will predict potential outcomes. We concatenate the separate representation $Z_{tre} = \varphi(x_{tre}, \vartheta)$ and $Z_{con} = \varphi(x_{con}, \vartheta)$ respectively with common representation $Z_{com} = \varphi(x, \vartheta)$ to amplify input features. Then we will get $\tau(x_{tre}, \vartheta) = \left[\varphi(x_{tre}, \vartheta), \varphi(x, \vartheta)\right]$ and $\tau(x_{con}, \vartheta) = \left[\varphi(x_{con}, \vartheta), \varphi(x, \vartheta)\right]$. This combination helps our model effectively extracts the shared features and preserves the unique characteristics of the specific representation.

We then use a feed-forward neural network $f(\cdot)$ to establish the outcome prediction network. We input the combination of representation $\tau(x_i, \vartheta)$ and treatment assignment t_i to predict potential outcome $f(\tau(x_i, \vartheta), t_i)$. We assess the performance by calculating the mean squared error between the predicted results and the ground truth observations Y_i. The loss function is defined as:

$$L_{pre} = \frac{1}{n} \sum_{i=1}^{n} (f(\varphi(x_i, \vartheta), t_i) - Y_i)^2 \tag{9}$$

4.5. Total loss and optimization

The ultimate objective function of this model integrates all components mentioned above, which is defined as:

$$\mathcal{L} = \alpha \mathcal{L}_{pre} + \beta \mathcal{L}_{diff} + \gamma \mathcal{L}_{recon} + \lambda \|W\|_2 \tag{10}$$

where $\mathcal{L}_{pre}$ means the loss of outcome prediction, $\mathcal{L}_{diff}$ means the loss of denoising for balanced representation, $\mathcal{L}_{recon}$ means the loss of representation and reconstruction. $\|W\|_2$ means l_2 regularization, which serves as a penalty term [49] to prevent overfitting. We employ the AdamW to optimize the parameters. AdamW combines the advantage of adaptive learning rate with weight decay.

α, β, γ, λ are hyper-parameters we designed to adjust the weights of different loss functions. This is a multi-objective optimization problem, so we assign weights of each part within a pre-specified range according to their relative importance. We set $\alpha > \beta > \gamma > \lambda$ during the training process to ensure the accuracy of outcome prediction and balanced representation of debiased distribution.

4.6. Theoretical analysis

We analyze the theoretical feasibility of our DBRT method by deriving an upper bound on the generalization error. $\{\mathcal{X} \to \mathcal{Z}, \mathcal{Z} \subset \mathbb{R}^v\}$ describes the process of mapping from the original input X to the latent space under treatment T and then get the prediction $Y(t)$. $\mathcal{H} \subseteq \{h : \mathcal{Z} \to \mathcal{Y}\}$ denotes the set of hypotheses learned by DBRT, where $f_t(x) = h_t \in \mathcal{H}$ for $t \in \{0, 1\}$. μ, ν are the distribution of the treated and control group. We get the expected error between the hypotheses and prediction function as $\mathbb{E}_{X,Y}[L(f_t(x), Y(t)) \mid X = x]$. We expand the formula and find its upper bound with reference to [6] for justification.

The dual representation of the 1-Wasserstein distance is listed as follows according to the Kantorovich–Rubinstein theorem,

$$W_1(\mu, \nu) = \sup_{\|l\| \le 1} \mathbb{E}_{\mu\sim\mu}[L(u)] - \mathbb{E}_{\nu\sim\nu}[L(v)] \tag{11}$$

where $\|l\|$ is the Lipschitz seminorm and can be written as $\sup \frac{|l(u)-l(v)|}{\rho(u,v)}$. ρ is the distance function between u ad v. Combined with the mean value theorem and the local Lipschitz condition, we can get the bound

as follows:

$$
\begin{aligned}
&\mathbb{E}_{X,Y}[L(f_t(x), Y(t)) \mid X = x] = \mathbb{E}_{X,Y}[(f_t(x) - Y(t))^2] \\
&= \mathbb{E}_{v\in v}[(f_t(x) - f_t^*(x) + f_t^*(x) - Y(t))^2] \\
&\le 2\mathbb{E}_{v\in v}[\left(f_t^*(x) - Y(t)\right)^2] + 2\mathbb{E}_{v\in v}[\left(f_t(x) - f_t^*(x)\right)^2] \\
&= 2\mathbb{E}_{\nu}[L(f_t^*(x), Y(t))] + 2\mathbb{E}_{\mu}[L(f_t(x), f_t^*(x))] + 2(\mathbb{E}_{\nu}[L(f_t(x), f_t^*(x))] \\
&- \mathbb{E}_{\mu}[L(f_t(x), f_t^*(x))]) \\
&\le 2\mathbb{E}_{\nu}[L(f_t^*(x), Y(t))] + 2\mathbb{E}_{\mu}[L(f_t(x), f_t^*(x))] \\
&+ 2 \sup_{\|l\|\le\kappa} \mathbb{E}_{\nu}[L(f_t(x), f_t^*(x))] - \mathbb{E}_{\mu}[L(f_t(x), f_t^*(x))] \\
&= 2\mathbb{E}_{\nu}[L(f_t^*(x), Y(t))] + 2\mathbb{E}_{\mu}[(f_t(x) - f_t^*(x))^2] + 2\kappa W_1(\mu, \nu) \\
&= 2\mathbb{E}_{\nu}[L(f_t^*(x), Y(t))] + 2\mathbb{E}_{\mu}[(f_t(x) - Y(t) + Y(t) - f_t^*(x))^2] + 2\kappa W_1(\mu, \nu) \\
&\le 2\mathbb{E}_{\nu}[L(f_t^*(x), Y(t))] + 2[2\mathbb{E}_{\mu}[\left(f_t(x) - Y(t)\right)^2] + 2\mathbb{E}_{\mu}[\left(f_t^*(x) - Y(t)\right)^2]] \\
&+ 2\kappa W_1(\mu, \nu) \\
&= 4\mathbb{E}_{\mu}[L(f_t(x), Y(t))] + 2\kappa W_1(\mu, \nu) + \eta.
\end{aligned}
\tag{12}
$$

where $\eta = 2\mathbb{E}_{\nu}[L(f_t^*(x), Y(t))] + 4\mathbb{E}_{\mu}[L(f_t^*(x), Y(t))]$, $f_t^*(x)$ is set as the optimal hypothesis where shift can be debiased. With the proof of the error bound, it demonstrates the generalization and credibility of our proposed DBRT.

5. Experiments

5.1. Datasets

To evaluate the performance of our model in different scenarios, we conduct experiments in several datasets for individual treatment effect estimation. While it is challenging to evaluate in common observational datasets due to the lack of counterfactual outcomes. So we follow previous work [17] and employ the public datasets: IHDP, Jobs, Twins.

IHDP: The Infant Health and Development dataset [50] was created from randomized controlled trials to assess the impact of childcare on preterm infants with low birth weight. Covariates describe the characteristics of the infants and their mothers, including birth weight, neonatal health index, mother's age and so on. If the infants receive high-quality childcare such as expert home visits, it belongs to the treated group and the control group otherwise. The outcome, as the cognitive score of infants, is simulated from setting "A" in the NPCI package [16]. Then there are 747 units with 25 covariates in this set.

Twins: The Twins dataset was obtained from the real data in America between 1989 and 1991 [51]. There are 114 000 pairs of twins in the dataset with the same sex and weighing less than 2000 g. After removing missing features, there are 30 covariates describing the conditions of pregnancy and parents, such as health, race, educational background and so on. The heavier one is considered as in the treated group, while the lighter is in the control group as [10]. The mortality after one year is regarded as the outcome. We will lead the procedure of $t|x \sim Bern(Sigmoid(w^T x + n))$ where $w^T \sim \mathcal{U}((-0.1, 0.1)^{30\times1})$ and $n \sim \mathcal{N}(0, 0.1)$ to the datasets as [16] so that selection bias can be simulated.

Jobs: The Jobs dataset is a combination of randomized controlled trials on the National Supported Work program and observational data [52,53]. It aims to evaluate the effect of vocational training on employment status with 8 covariates. The treated group denotes receiving job training, while the control group denotes not. There are 297 treated units and 425 control units in the randomized controlled trials part of the datasets, as well as 2490 control units in the observational data part. So we can get average treatment effect on the treated (ATT) from Jobs dataset [16]. ATT is defined as:

$$
ATT = \frac{1}{|T_1 \cap E|} \sum_{x_i \in T_1 \cap E} Y_1(x_i) - \frac{1}{|T_0 \cap E|} \sum_{x_i \in C \cap E} Y_0(x_i) \tag{13}
$$

where T_1 denotes the subset of treated units, T_1 denotes the subset of control units, E denotes the subset of all units from randomized controlled trials.

Table 1
Search spaces of hyper-parameters.

Hyper-parameter	Range
batch_size	32, 64, 128, 256
Learning rate	$1e{-}2$, $1e{-}3$, $1e{-}4$, $1e{-}5$
Depth of predictor layers	1, 2, 3, 4, 5, 6
Depth of decoder layers	1, 2, 3, 4, 5, 6
n_{head}	2, 3, 4
Time-step	4, 8, 12, 16, 20, 25, 30, 35, 40
Optimizer	RMSProp, Adam, AdamW
α	0.2, 0.4, 0.8, 1.0, 2.0, 4.0, 10.0
β	0.2, 0.4, 0.8, 1.0, 2.0
γ	0.2, 0.4, 0.8, 1.0, 2.0

5.2. Evaluation metrics

We use ϵ_{ATE} and $\sqrt{\epsilon_{PEHE}}$ [50] as metrics to evaluate IHDP and Twins, which respectively calculating mean absolute error on average treatment effect (ATE) and the rooted Precision in Estimation of Heterogeneous Effects (PEHE). ϵ_{ATE} and $\sqrt{\epsilon_{PEHE}}$ can be defined as follows:

$$
\epsilon_{ATE} = \frac{1}{n}\left|\boxed{\sum_{i=1}^{n}(Y_i^{t=1} - Y_i^{t=0})} - \sum_{i=1}^{n}(\hat{Y}_i^{t=1} - \hat{Y}_i^{t=0})\right| \tag{14}
$$

$$
\sqrt{\epsilon_{PEHE}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}((Y_i^{t=1} - Y_i^{t=0}) - (\hat{Y}_i^{t=1} - \hat{Y}_i^{t=0}))^2} \tag{15}
$$

where $Y_i^{t=1}$, $Y_i^{t=0}$ are the ground truth of treated and control group, $\hat{Y}_i^{t=1}$, $\hat{Y}_i^{t=0}$ are their estimations from our model.

Considering the lack of ground truth on Jobs dataset, we use ϵ_{ATT} and policy risk $R_{pol}(\pi)$ as evaluation metrics. ϵ_{ATT} evaluates ATT in terms of mean absolute error, which is similar to ϵ_{ATE} on treated samples. ϵ_{ATT} can be defined as:

$$
\epsilon_{ATT} = |ATT - \frac{1}{|T_1 \cap E|} \sum_{x_i \in T_1 \cap E} \hat{Y}_1(x_i) - \hat{Y}_0(x_i)| \tag{16}
$$

where $\widehat{Y}_1$ and $\widehat{Y}_0$ denote the predicted treated and control outcome. Policy risk calculates the average loss in value when treating based on the policy suggested by the estimator [16]. It is defined as:

$$
\begin{aligned}
R_{pol}(\pi) = 1 - [&\mathbb{E}(Y_i^{t_i=1}|\pi(x_i) = 1) \cdot P(\pi(x_i) = 1) \\
&+ \mathbb{E}(Y_i^{t_i=0}|\pi(x_i) = 0) \cdot P(\pi(x_i) = 0)]
\end{aligned}
\tag{17}
$$

when $\hat{Y}_i^{t_i=1} - \hat{Y}_i^{t_i=0} > 0$, $\pi(x_i) = 1$ means the policy in treated condition and $\pi(x_i) = 0$ otherwise.

Each dataset is split into training/validation/testing sets with specific proportions as [17]. We run over 100 times on each dataset and then average them as the experiment result. When we evaluate performance of the model, each dataset is divided into the case of in-sample and out-of-sample [16], corresponding to the results of the training and testing sets.

5.3. Training detail

Due to the absence of counterfactual outcomes in the observational dataset, the standard cross-validation approach is not well-suited for this situation. Therefore, our study follows the methodologies employed in [32] and [16], and seek the optimal within the hyperparameter space based on the validation set results. The specific search spaces are provided in Table 1. We first identify an optimal parameter and then fix its value while searching for the next parameter.

Time-step is the number of step to predict and remove the noise which we designed for the denoising for balanced representation. α, β, γ are the weights for $\mathcal{L}_{pre}$, $\mathcal{L}_{diff}$ and $\mathcal{L}_{recon}$ respectively. n_{head} is the number of head which we designed for the Memory-Augmented Attention encoder.

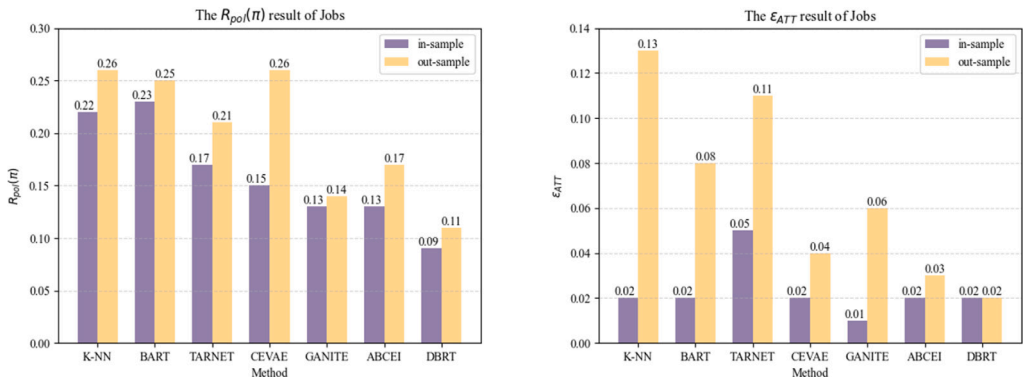

Fig. 2. Performance comparison of $R_{pol}(\pi)$ and ϵ_{ATT} on Jobs dataset, where closer to the bottom-line denotes better performance.

Table 2
Optimal hyper-parameters of DBRT.

Dataset	Optimal hyper-parameters
IHDP	Batch size: 64, time-step: 30, α: 1.0, β: 0.4, γ:0.2
Jobs	Batch size: 64, time-step: 12, α: 1.0, β: 0.2, γ:0.2
Twins	Batch size: 128, time-step: 30, α: 1.0, β: 0.4, γ:0.2

We further improve the model performance by optimizing the hyperparameters listed above. After conducting extensive experiments on all datasets, we have obtained a set of consistent parameters. For example, the learning rate is set to $1e-3$, both the depth of predictor and decoder layers is set to 3, n_{head} is set to 3, and AdamW is set as the optimizer. The remaining specific parameter configurations are listed in Table 2.

5.4. Experimental results

In this part, we employ the proposed DBRT method to estimate individual causal effect and evaluate its performance on different datasets using corresponding metrics. We compare the obtained results with those of classical baseline methods. Baseline methods include least squares regression using treatment as a feature (OLS/LR1); separate least squares regressions for each treatment (OLS/LR2); balancing linear regression (BLR) [25]; k-nearest neighbor (k-NN) [54]; Bayesian additive regression trees (BART) [23]; balancing neural network (BNN) [25]; treatment-agnostic representation network (TARNET) [16]; local similarity preserved individual treatment effect (SITE) [17]; causal effect variational autoencoders (CEVAE) [10]; Adversarial balancing-based representation learning for causal effect inference (ABCEI) [32]; Generative Adversarial Nets for inference of individualized treatment effects (GANITE) [11]; Direct learning with multi-task neural networks for treatment effect estimation (HTENet) [40]; Counterfactual Inference with Hidden Confounders Using Implicit Generative Models (CEIGM) [29]; Cycle-Balanced Representation Learning For Counterfactual Inference (CBRE) [33]; Contrastive Individual Treatment Effects Estimation (CITE) [27]; Advancing Counterfactual Inference through Quantile Regression (ACIQR) [41]. We will then illustrate and analyze the experimental results as follows.

Table 3 and Fig. 2 show the comparison of our model DBRT with baseline methods on three datasets of Twins, Jobs, IHDP. Given that the performance divergence among different methods on the Jobs dataset are not significant, we employ bar charts to intuitively compare. It is obvious that our algorithm outperforms previous state-of-the-art methods generally. While the in-sample results of (ϵ_{ATT}) on Jobs are not the best, there is an improvement on the out-of-sample and $R_{pol}(\pi)$ performance. Although ACIQR and CBRE has shown competitive performance, we still manage to surpass it.

We have applied traditional machine learning techniques and conducted comparative evaluations. Taking the methods shown in the table as example, regression-based methods such as OLS/LR1 and OLS/LR2, matching methods like K-NN and tree-based methods like BART have been proposed to solve the issue. But there are limitations that they often assume the training and test data are drawn from the same distribution. In practice, confounders in observational data leads to distribution shift. So these methods may struggle to handle confounding variables and distribution bias, resulting in inferior performance compared to representation learning models such as TARNET in general. It demonstrates that removing bias in observational data is necessary for the accuracy of outcome prediction in practice.

Representation learning models, such as TARNET, BNN and SITE, design corresponding structures to handle the complexities induced by confounders and distribution bias. Their ability to leverage features and construct balanced representations in latent space allows them to effectively model and address these challenges, leading to great progress in performance. In particular, it also demonstrates that the SITE model has better performance by taking local similarity information of covariates into account.

Among the generative methods, CEVAE performs less favorably. We suppose that VAE method will encode the latent variables as Gaussian distribution, which is likely to introduce extra noise in our task. The result of ABCEI and GANITE demonstrates the superiority of GAN-based methods. While, our approach's evaluation with both lower error and variance also demonstrates its high stability.

Recent work has made further improvements compared to previous classical methods, achieving varying degrees of performance enhancement. Among them, ACIQR demonstrate impressive performance on the IHDP dataset, particularly on the training set, which is comparable to our model. While, our model consistently outperforms ACIQR on the test set. ACIQR achieves counterfactual inference through nonlinear quantile regression with the advantage of conducting causal inference based solely on observational data without the need for predefined causal model. Moreover, ACIQR employs neural networks within a bi-level optimization scheme,enabling it to handle diverse data structures. However, the identifiability of counterfactual outcomes remains a challenge in scenarios involving non-monotonic generative functions. While our model surpasses ACIQR in tabular data, ACIQR is also applicable to image tasks. We will extend our method in future study.

HTENet employs a multi-task neural network architecture designed for direct learning to estimate treatment effects. This end-to-end approach avoids model misspecification and high variability issues inherent in traditional two-stage methods. However HTENet may encounter challenges with data imbalance, resulting in suboptimal results. CITE innovatively integrates the self-supervised information latent within the data to generate balanced representations. It outperforms the previously mentioned classical representation learning methods. CBRE, as an optimized version of ABCEI, outperforms ABCEI in experimental result, which proves the efficacy of cycle consistency.

Table 3
Performance comparison of $\sqrt{\epsilon_{PEHE}}$ and ϵ_{ATE} on IHDP and Twins datasets. Both evaluations are the lower the better. Bold indicates the best performance. In-sample column denotes the result for training set; Out-sample column denotes the result for test set.

Method	IHDP ($\sqrt{\epsilon_{PEHE}}$)		IHDP (ϵ_{ATE})		Twins ($\sqrt{\epsilon_{PEHE}}$)		Twins (ϵ_{ATE})	
	In-sample	Out-sample	In-sample	Out-sample	In-sample	Out-sample	In-sample	Out-sample
OLS/LR$_1$	5.8 ± .3	5.8 ± .3	.73 ± .04	.94 ± .06	.319 ± .001	.318 ± .001	.004 ± .003	.007 ± .006
OLS/LR$_2$	2.4 ± .1	2.5 ± .1	.14 ± .01	.31 ± .02	.320 ± .002	.320 ± .003	.004 ± .003	.007 ± .006
K-NN	2.1 ± .1	4.1 ± .2	.14 ± .01	.79 ± .05	.333 ± .001	.345 ± .007	.003 ± .002	.005 ± .004
BART	2.1 ± .1	2.3 ± .1	.23 ± .01	.34 ± .02	.347 ± .009	.338 ± .016	.121 ± .024	.127 ± .024
BLR	5.8 ± .3	5.8 ± .3	.72 ± .04	.93 ± .05	.312 ± .003	.323 ± .018	.006 ± .004	.033 ± .009
BNN	2.2 ± .1	2.1 ± .1	.37 ± .03	.42 ± .03	.331 ± .004	.337 ± .008	.006 ± .003	.020 ± .007
TARNET	.88 ± .02	.95 ± .02	.26 ± .01	.28 ± .01	.317 ± .002	.315 ± .003	.011 ± .002	.015 ± .002
SITE	.67 ± .02	.71 ± .04	.22 ± .01	.24 ± .01	.309 ± .002	.311 ± .004	.016 ± .001	.020 ± .002
CEVAE	2.7 ± .1	2.6 ± .1	.34 ± .01	.46 ± .02	.359 ± .001	.367 ± .003	.022 ± .002	.032 ± .003
ABCEI	.71 ± .0	.73 ± .0	.09 ± .01	.09 ± .01	.294 ± .002	.302 ± .004	.003 ± .001	.005 ± .001
GANITE	1.9 ± .4	2.4 ± .4	.23 ± .01	.24 ± .03	.289 ± .005	.297 ± .016	.004 ± .001	.005 ± .001
HTENet	1.02 ± .94	1.18 ± 1.27	.21 ± .13	.24 ± .22	.382 ± .012	.386 ± .011	.014 ± .003	.017 ± .010
CEIGM	2.0 ± .1	2.0 ± .2	1.1. ± .2	1.2 ± .2	.476 ± .010	.503 ± .006	.010 ± .004	.017 ± .003
CBRE	.51 ± .03	.60 ± .12	.10 ± .02	.14 ± .01	.290 ± .003	.294 ± .010	.004 ± .003	.005 ± .008
CITE	.58 ± .1	.60 ± .1	.09 ± .01	.11 ± .02	.312 ± .001	.320 ± .003	.004 ± .002	.004 ± .006
ACIQR	.39 ± .03	.47 ± .1	.06 ± .03	.09 ± .04	.290 ± .003	.292 ± .003	.003 ± .002	.004 ± .001
DBRT (Ours)	**.39 ± .02**	**.44 ± .03**	**.05 ± .01**	**.06 ± .02**	**.267 ± .002**	**.271 ± .005**	**.002 ± .001**	**.003 ± .001**

Table 4
Ablation study. $\mathcal{L}_{recon}$ denotes the module with memory-augmented encoder to map the data into latent space, $\mathcal{L}_{diff}$ denotes the module with denoising network for balanced representation, $\mathcal{L}_{HSIC}$ denotes HSIC as the metric for computing the reconstruction loss. Twins and IHDP are evaluated with $\sqrt{\epsilon_{PEHE}}$, Jobs is evaluated with $R_{pol}(\pi)$. Both are the lower the better.

	Twins ($\sqrt{\epsilon_{PEHE}}$)		Jobs ($R_{pol}(\pi)$)		IHDP ($\sqrt{\epsilon_{PEHE}}$)	
	In-sample	Out-sample	In-sample	Out-sample	In-sample	Out-sample
$\mathcal{L}_{recon}$	.380 ± .003	.378 ± .004	.28 ± .02	.29 ± .03	1.48 ± .07	1.51 ± .12
$\mathcal{L}_{diff}$	.344 ± .004	.347 ± .002	.14 ± .02	.17 ± .02	.61 ± .01	.72 ± .05
$\mathcal{L}_{HSIC}$	.381 ± .004	.381 ± .002	.24 ± .00	.27 ± .02	1.21 ± .09	1.23 ± .08
$\mathcal{L}_{recon}+\mathcal{L}_{diff}$	.273 ± .004	.280 ± .005	.14 ± .01	.16 ± .01	.51 ± .03	.55 ± .01
$\mathcal{L}_{recon}+\mathcal{L}_{HSIC}$	.357 ± .007	.377 ± .009	.17 ± .03	.21 ± .03	1.17 ± .01	1.22 ± .03
$\mathcal{L}_{diff}+\mathcal{L}_{HSIC}$	.301 ± .004	.309 ± .007	.14 ± .03	.17 ± .02	.42 ± .03	.44 ± .04
$DBRT$	**.267 ± .002**	**.271 ± .005**	**.09 ± .02**	**.11 ± .04**	**.39 ± .02**	**.44 ± .03**

In summary, it is evident that regardless of whether classic methods mentioned or recent work are considered, our model achieves state-of-the-art performance, which demonstrates the effectiveness of our methodology.

5.5. Ablation study

We conduct ablation experiments to demonstrate the importance of each proposed module in our model. By keeping the specific module and removing other parts, we assess the module's contribution to the results. As shown in Table 4, we first conduct experiments by retaining only one module. $\mathcal{L}_{diff}$ surpasses the others and even outperforms the combination of $\mathcal{L}_{recon}$ and $\mathcal{L}_{HSIC}$, which demonstrates the effectiveness of the proposed denoising method in achieving balance in representation.

In general, the results of combining two modules are superior to those obtained from a single module, particularly when $\mathcal{L}_{diff}$ combined with the other two modules. The row of $\mathcal{L}_{recon}+\mathcal{L}_{diff}$ exhibits the best performance, which proves the benefit of extracting relationships between features for better representation. $\mathcal{L}_{diff}+\mathcal{L}_{HSIC}$ denotes that we input original covariates into denoising module. The change in IHDP is relatively small and even equal in out-sample, presumably due to the fewer features in this dataset. This module may be useful when addressing intricate relationships of numerous covariates. We employ mean square error for reconstruction evaluation instead in the $\mathcal{L}_{recon}+\mathcal{L}_{diff}$ row. Compared to the final results of DBRT, this part also exhibits a slight decrease in performance. It highlights the suitability of HSIC for measuring relationship after non-linear transformations like attention mechanism.

In conclusion, our ablation experiments demonstrate the significance of each module in improving the model's performance, which denoising for balanced representation makes the most contribution. It is further affirmed that incorporating all the proposed modules together would finally achieve the best performance.

5.6. Representation visualization

We employ t-SNE to visualize the original data and the variation in representations across different time-steps for the IHDP and Jobs datasets as a contrast. In this visualization, we focus on the relative position of the points and disregard the shape they formed. As the similarity of data distributions increases, the distances between data points tend to decrease.

As Fig. 3 shows, the points of both the treated and control group concentrate within their respective clusters, suggesting the shared underlying factors among covariates with the same treatment assignment. For the original inputs of the two datasets, there is obvious difference between the two groups, confirming the presence of distribution shift.

After applying our model's representations and denoising on the fifth timestep, the overlap between the two distributions begins to occur, with the Jobs dataset exhibiting a higher degree of overlap. This can be attributed to the fact that the optimal hyperparameter of timestep for the Jobs dataset is smaller than that of IHDP, as mentioned in Section 5.3. Consequently, the two distributions tend to reach balance earlier.

For the final representation of DBRT, we can see a significant overlap between the distributions of the treated and control groups. The representation of treated group is almost embedded within the control because of its relatively smaller size, which highlights good distribution balance. The visual comparisons effectively demonstrate the essential role of our model in achieving representation balance.

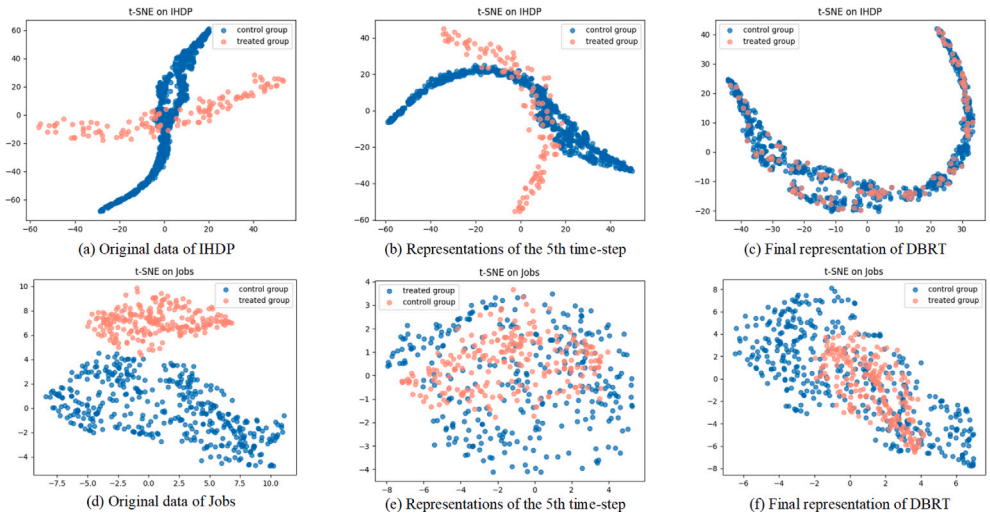

Fig. 3. t-SNE visualization on IHDP and Jobs. Pink circles denote the distribution of treated, blue circles denote the distribution of control. The left is the original data, the middle is the representation of the 5th timestep in DBRT, the right is the final representation of DBRT.

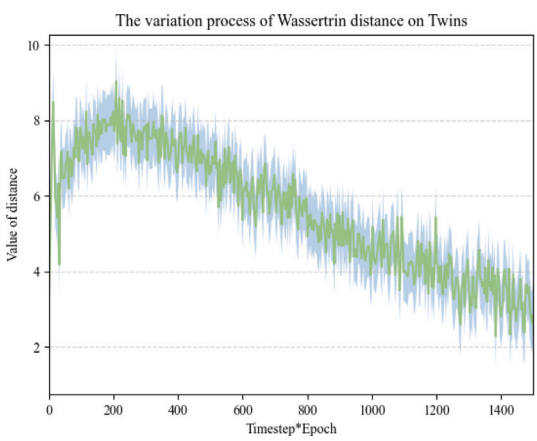

Fig. 4. The variation process of Wasserstein distance on Twins dataset. It corresponds to the Wasserstein distance between the treated and control groups across 30 timesteps under 50 epochs. The shadow denotes the variance.

While t-SNE visualization is not intuitive on the Twins dataset due to the large sample size. So we visualize the Wasserstein distance between the treated and control groups instead. As shown in Fig. 4, we present the variation process across 30 timesteps under 50 epochs. The observed fluctuation in the variation process can be attributed to the decreasing trend within each epoch, while the results tend to be relatively larger in the initial stage of the subsequent epoch. Overall, the trend exhibits a declining pattern and tends to converge, which indicates the distribution shifts can be addressed by the proposed DBRT.

5.7. Network for denoising

As mentioned above, we construct a specific network with simplified U-net to predict the bias between the treated and control group. In this part, we replace the U-net with two additional network structures to explore the impact of the simplified U-net model on the model performance. We employ MLP network and ResNet as the comparisons. We set them both 3-layer which is the same as the simplified U-net to ensure a fair comparison. Table 5 shows the performance results of different denoising network on datasets.

We notice that our simplified U-net has a slight improvement in almost all of the results. While on IHDP dataset, the results are comparable and even worse. We suppose that the network may still be incapable to capture sufficient information from a limited number of covariates, leading to a decrease in performance. In this condition, the simple MLP network is ideal to capture the discrepancies between the treated and control group, particularly in scenarios with fewer covariates. ResNet and U-net perform better on datasets with a larger number of covariates, such as Twins and Jobs datasets. They both employ skip connections to reduce information loss. However, U-net, with its symmetric encoder and decoder structure, has the ability to capture more detailed information and needs less parameters. After our simplified design in U-net's layer and dimension, it is well-suited for our task to learn the divergence between the two distribution and will excel in dealing with intricate relationship of data.

5.8. Memory-vector in attention

To further demonstrate the effectiveness of adding memory-vectors to the attention mechanism in our scenario, we conducted ablation experiments on the feature extraction network. We compared the performance of the attention mechanism with and without memory-vectors, as well as the approach of directly inputting original features into the subsequent denoising module without using a feature extraction layer.

The results indicate that attention mechanisms consistently enhance performance over raw feature inputs. As shown in Table 6, adding memory vectors to the attention mechanism improves outcomes across various datasets, particularly on the Twins dataset. We attribute this to its larger sample size, allowing memory vectors to better capture feature interactions. In future work, we can explore applying our method in more complex scenarios.

6. Conclusion

In this study, we consider the fundamental cause of selection bias and then propose a denoising approach to achieve balanced representation for individual treatment effect estimation. Additionally, we employ Memory-Augmented Attention to map covariates into appropriate latent space, with the constraints of HSIC to enhance the representations of original data. By leveraging these innovations, our model not only mitigates the risk of distribution shift but also captures intricate relationships within the data more accurately. We conduct ample experiments on three datasets and compare our approach with previous classical methods. It shows that our method outperforms the baselines

Table 5
Performance comparison of U-net with other network, Twins and IHDP are evaluated with $\sqrt{\epsilon_{PEHE}}$, Jobs is evaluated with $R_{pol}(\pi)$. Both are the lower the better.

Model	Twins ($\sqrt{\epsilon_{PEHE}}$)		Jobs ($R_{pol}(\pi)$)		IHDP ($\sqrt{\epsilon_{PEHE}}$)	
	In-sample	Out-sample	In-sample	Out-sample	In-sample	Out-sample
U-net	**.267 ± .002**	**.271 ± .005**	**.09 ± .01**	**.11 ± .04**	.39 ± .02	.44 ± .03
MLP	.263 ± .002	.287 ± .002	.11 ± .02	.12 ± .06	**.37 ± .04**	**.44 ± .02**
ResNet	.261 ± .003	.283 ± .004	.10 ± .02	.12 ± .01	.40 ± .02	.45 ± .02

Table 6
Performance comparison of attention mechanism with and without memory vector, Twins and IHDP are evaluated with $\sqrt{\epsilon_{PEHE}}$, Jobs is evaluated with $R_{pol}(\pi)$. Both are the lower the better.

Model	Twins ($\sqrt{\epsilon_{PEHE}}$)		Jobs ($R_{pol}(\pi)$)		IHDP ($\sqrt{\epsilon_{PEHE}}$)	
	In-sample	Out-sample	In-sample	Out-sample	In-sample	Out-sample
With-Mem	**.267 ± .002**	**.271 ± .005**	**.09 ± .01**	**.11 ± .04**	**.39 ± .02**	**.44 ± .03**
Without-Mem	.275 ± .001	.287 ± .001	.09 ± .02	.12 ± .01	.41 ± .02	.46 ± .04
MLP	.301 ± .004	.309 ± .007	.14 ± .03	.17 ± .02	.42 ± .03	.44 ± .04

in general. Furthermore, we visualize the representation with T-SNE, which reveals a significant reduction in distribution bias after applying our model. Given the above, it validates the remarkable efficacy of our methodology in debias distribution and effect estimation.

In future work, we will further our analysis beyond binary treatment by exploring other types of data, such as multiple and continuous treatment. We plan to design corresponding decoders to address different forms of treatment variables. Then it will provide stronger support for decision-making. We also notice that contrasting learning helps explore the hidden confounder in covariates [27,55]. So we may introduce this method to our model in the next step for more novel analysis.

CRediT authorship contribution statement

Hai Yang: Supervision, Project administration, Investigation, Funding acquisition. **Jing Yao:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Zhe Wang:** Supervision, Resources, Project administration, Funding acquisition. **Yijing Yang:** Validation, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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